Tactile-Based Fabric Defect Detection Using Convolutional Neural Network With Attention Mechanism

Bin Fang, *Member, IEEE*, Xingming Long, Fuchun Sun, *Fellow, IEEE*, Huaping Liu, *Senior Member, IEEE*, Shixin Zhang, and Cheng Fang, *Member, IEEE*

***Abstract*— This article proposes a fabric structure defect detection method based on the vision-based tactile sensor. The result will be robust by using the tactile sensor regardless of dyeing patterns which can influence the result if some other sensors are used, e.g., vision perception. It also reduces the influence of ambient light on defect detection. Therefore, the proposed method can be more robust and universal than conventional visual methods. A robotic arm equipped with the tactile sensors was used to automate and standardize the data collection process and construct fabric datasets. In addition, a convolutional neural network (CNN) integrated with attention mechanism in the channel domain was developed to detect fabric types. The proposed network employed frequency domain filtering to remove or weaken the influence of normal fabric texture information to improve defect detection efficiency and accuracy. Finally, several experiments were conducted to demonstrate the proposed method's superiority to a visual defect detection method for detecting structural defects. In addition, the efficiency of the proposed method is evaluated. Experimental results show that the proposed method is feasible and efficient to meet the real-world detection requirements.**

***Index Terms*— Attention mechanism, defect detection, vision-based tactile sensor.**

I. Introduction

FABRIC is one of the most essential products in modern life. Many of our daily necessities are directly or indirectly produced from fabric, such as clothes, towels, and bandages. Therefore, the production of fabric is one of the most important industries in the world. However, in the fabric production process, due to machine faults or system errors, various fabric defects are produced, such as missing yarns. The value of fabric would decrease by approximately 50% if it has some defects [1]. As a result, fabric defect detection is indispensable in the fabric production process [2], [3]. Furthermore, it is crucial to develop an automatic fabric defect detection method [4], [5] with high robustness and accuracy, which can significantly improve the production efficiency and quality of fabric in factories [6].

Current fabric defect detection algorithms are developed using fabric images captured by cameras [7]. By setting up a camera above the fabric production line, images captured by the camera are used to automatically detect fabric defects. Based on existing image processing methods, fabric defect detection methods can be divided into two categories.

The first is traditional image processing methods, which are widely used in fabric defect detection. These traditional image processing methods are mainly divided into three categories: statistics-based methods, frequency-based methods, and model-based methods. First, statistics-based methods extract artificially defined features that can be compared to determine the difference between an image of defective fabric and that of normal fabric. This method typically extracts features from a subimage. If the feature value deviates from the mean value of the corresponding feature of the entire fabric image, the subimage contains defects. At present, different feature choices for fabric defect detection exist. For example, Selver *et al.* [8] used histogram features, and Sadaghiyanfam [9] and Kaynar *et al.* [10] analyzed gray-level co-occurrence matrix and local binary pattern features. Bu *et al.* [11] and Hanmandlu *et al.* [12] used fractal dimensions as features. Second, frequency-based methods use the frequency-domain transformation of an image to detect fabric defects. Becafuse fabric images often have rich texture features, and the frequency of defects is different from that of the surrounding texture features, fabric image features in the frequency domain can be used for defect detection. Image processing methods based on the frequency domain mainly include the fast Fourier transform, discrete cosine transform, wavelet transform, and Gabor transform [13]. Fast Fourier transform and discrete cosine transform are used to preserve defect information using a high-pass filter after image transformation. Then, inverse transformation is used to restore the transformed image and perform other operations, which is equivalent to image preprocessing. The wavelet transform [14]

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and Gabor transform [15] can be used as the main parts of the defect detection algorithm to directly detect the defect location using frequency domain information because they can obtain position information about the frequency. Third, model-based methods mainly include modeling and decomposition methods [16]. Among all model-based methods, the low-rank decomposition method is the most commonly used [17], [18]. Because texture information about fabric is highly autocorrelated, the texture component of a fabric image can be separated via a low-rank decomposition, obtaining the defect component of the fabric image.

The second class of methods is based on machine learning algorithms [19]. Among them, deep convolutional neural networks (CNNs) have been widely used in image feature extraction because of their powerful feature extraction ability. They have achieved superior performance compared to traditional feature extraction methods. Nowadays CNNs have been widely used in image-based defect detection [20]–[22]. In fabric defect detection, many researchers have conducted related studies on deep learning [23], [24]. The results have verified that CNNs can be effectively used to extract fabric images features and can better detect fabric defects than other methods.

Generally, the defects of fabric are divided into two categories: dyeing defects referring to the appearance defects of a fabric caused by uneven dyeing or oil that leaves stains on the fabric; structural defects referring to knitting defects caused by missing yarns or broken holes. Structural defects are more damaging to fabric quality and difficult to detect than dyeing defects. In this article, we propose a new method for detecting structural defects in fabric using tactile information. It is an extension of our previous conference paper [25]. The designed vision-based tactile sensor is used to capture fine tactile images that can effectively avoid the influence of fabric texture or dyeing patterns on defect detection. Furthermore, it can minimize the influence of the environmental conditions (e.g., lighting conditions). These factors contribute to the difficulty and complexity of fabric defect detection with vision. For example, Wang *et al.* [26] first classified fabric according to its pattern and then trained a defect detection model for each fabric pattern type. This approach increased the calculation cost and could not detect defects in other fabrics with new patterns. The proposed tactile information-based method can avoid this problem and improve the generality of the detection model.

The main contributions of this article can be summarized as follows.

1) A new method for fabric defect detection based on tactile information is proposed and a fabric tactile dataset is established. The proposed method avoids the influence of fabric surface pattern and environment on detection algorithms, which can effectively improve the generality and robustness of the method. Experimental results show that the tactile-based detection method has a higher detection accuracy than the visual methods.
2) A CNN integrated with the attention mechanism is designed that can effectively improve the training stability and detection accuracy of tactile methods.
3) A novel evaluation method of detection efficiency is developed to prove that the proposed method is a feasible solution to the applications of tactile detection methods.

This article is divided as follows. Section II introduces the vision-based tactile sensor. Section III introduces the experimental platform and detection algorithm. Section IV introduces the experimental results and discusses the advantages of the tactile detection method. Finally, Section V concludes this article.

II. Vision-Based Tactile Sensor

Tactile sensation is an essential form of perception in humans. It helps us better recognize objects in different light environments and allows us to distinguish between objects of similar appearance (e.g., similar colors) that are made of different materials. Nowadays various tactile sensors have been developed for robotic perception, including resistance sensors [27], capacitive sensors [28], and piezoelectric sensors [29]. The most critical problem with these electrical tactile sensors is that high tactile resolutions can be hardly obtained so that they are unsuitable for fabric defect detection. Vision-based tactile sensors have high resolution and can obtain rich information from the object surface. Therefore, vision-based tactile sensors have attracted research interest recently [30]–[32].

Considering the requirements of fabric defect detection, we need to detect relatively subtle structural damages. Hence a vision-based tactile sensor is selected [33]. The structure of a vision-based tactile sensor consists mainly of a camera, a support layer, an elastomer, an attachment layer, and an LED. The camera is mounted at the bottom, and the transparent elastomer and transparent support structure used to fix the elastomer are located directly above the camera. The surface of the transparent elastomer has an attachment layer that can be deformed with the deformation of the transparent elastomer. Its main role is to prevent the influence of external light on the camera. In addition, it should have a good reflecting characteristic, so that it can reflect the tactile information about an object's surface when the elastomer is deformed. Typically, metal powder is used as the attachment layer. When the sensor comes into contact with an object, the elastomer deforms to fit the texture of the object's surface, and the attachment layer deforms accordingly. A camera mounted on the bottom captures the deformation and acquires tactile information about the surface.

III. Methods

A. Dataset

A robotic arm equipped with tactile sensors was used to collect tactile data of fabric. The acquisition platform is depicted in Fig. 1. The sensor was fixed to the end of the robotic arm, which was controlled to randomly sample fabric samples. The arm would randomly select a position of a fabric sample and then control the sensor to press the fabric sample vertically down with the same pressure to obtain the tactile image of the fabric sample.

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Fig. 1. Tactile data acquisition platform.

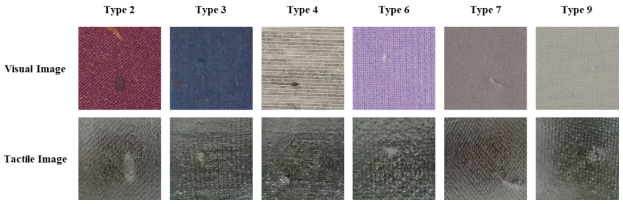


Fig. 2. Different types of fabric visual images and their corresponding tactile images.

Nine fabric types with different structural defects were used as samples for data. These fabrics contain various common patterns and fabric types, and each has different structural defects. Tactile images with and without defects are manually selected and labeled. Among them, images that do not contain any defects are selected as defect-free images. Images containing a defective area (including images containing only part of a defect) are selected as defective images. For each sample category, approximately 100 defective and 100 normal fabric images were selected, totaling 1768 tactile images.

In addition, fabric visual data was constructed for comparative experiments. A camera was set up above the fabric samples to collect their fabric images. Subsequently, the collected images were manually screened to remove the images captured on the edge of the fabric and then to determine whether each image contained defects. For each fabric sample, 100 defective and 100 normal fabric images were collected, totaling 1800 visual images.

Some tactile images and their corresponding visual images are shown in Fig. 2. The defects in the tactile images were clearer and these tactile images did not contain any dyeing information about the fabric. Furthermore, background texture information about the tactile image was more obvious than that in the visual image.

To further expand the dataset, data enhancement processing is performed. Each image was symmetrically processed to obtain new image data, thereby doubling the number of datasets is constructed.

No other preprocessing operations such as denoising and cropping are used on the dataset, and these images are directly used for subsequent training and testing.

B. Algorithm

When using tactile methods for fabric defect detection, the structural defect features of fabrics are more prominent, which is conducive to the feature extraction of a network model, and more subtle structural defects of the fabric can be detected. However, the textile texture characteristics of fabrics would also become more prominent, and these texture characteristics would affect the training process of the fabric defect detection model. During training, the texture characteristics of different fabrics increase the dimension of the overall fabric tactile image features, making the network more difficult to train and more prone to overfitting. Especially, deep network models have stronger image feature extraction capabilities, which can extract more feature information from the texture of fabrics. These features are invalid for defect detection, causing the training process to be easily influenced by the textures. In subsequent experiments, it was found that the more obvious texture features are in a tactile image, the greater the impact on the accuracy of the trained defect detection model.

The above-mentioned recognition noise comes from the textile texture of fabrics, and these textures often have obvious repetitiveness on a spatial scale, making the features of these textures in the frequency domain have obvious characteristics. Some conventional fabric defect detection algorithms, and frequency domain transformation methods have also been used to eliminate fabric texture information through filtering in the frequency domain [13]. When an image is retransformed to the spatial domain, some types of the fabric texture information can be eliminated or weakened, which can improve the accuracy of subsequent defect detection methods.

In CNNs, the convolution operations can also be regarded as frequency domain information extraction methods. Therefore, each convolution kernel is responsible for extracting specific frequency domain information in the previous feature map. After each convolution kernel is calculated, a channel in the feature map of the next layer will be correspondingly obtained. Therefore, in the calculation process of a CNN, each channel in the feature map of one layer corresponds to the information about a specific frequency domain in the feature map of the previous layer.

We consider using frequency domain filtering on the basis of CNNs to remove or weaken the influence of fabric texture information. Therefore, we hope to perform weighting calculations on various channels calculated in each layer of CNN and assign lower weights to channels corresponding to the frequency domain of the fabric texture to achieve an effect similar to frequency domain filtering.

Based on the above considerations, we use an attention mechanism based on the channel domain. The specific method is to use information about the feature map of the previous layer to calculate the weight value of each channel obtained in the next layer and perform the corresponding weighting calculation for each channel when outputting the feature map of the next layer. This enables the model to know which channels should be assigned low weights through training, that is, which frequency domains should be filtered.

In this study, the attention module based on the channel domain is added to the residual structure of the ResNet network, and the fabric defect detection model is obtained. Fig. 3 depicts the model structure of a model built on the Resnet50 network. The ResNet network's structure largely remains unchanged. The entire network is still divided into four parts

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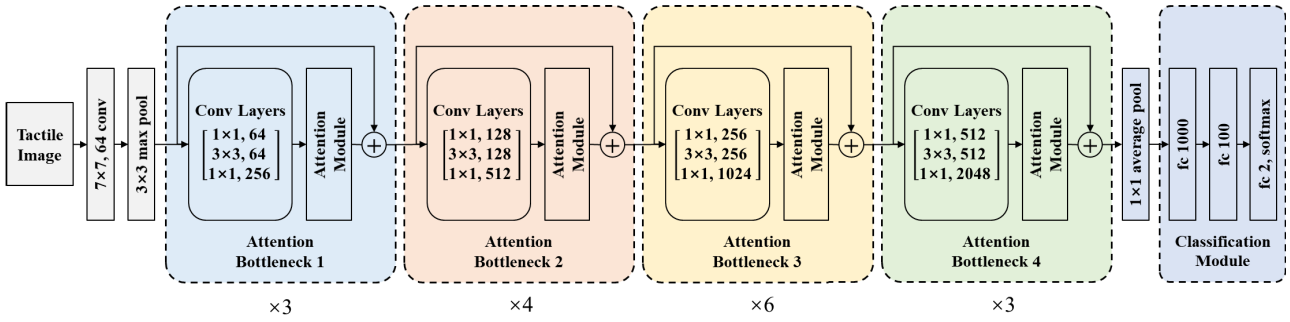


Fig. 3. Structure of a fabric defect detection model.

according to the number of output channels. In each part, the same bottleneck structure is repeatedly connected multiple times to obtain a deeper network structure. In the original ResNet's bottleneck structure, three convolution operations are included, and then the result of the convolution calculation is residually connected with the input of the bottleneck. In the attention-based ResNet structure, after three convolution operations, an attention module based on the channel domain is added to perform weight calculation and weighting operations on the convolution calculation results for each channel. Finally, the weighted result of the convolution calculation is residually connected with the input of the bottleneck. Placing the attention module before the residual connection is to ensure that the effect of the residual connection in ResNet is eliminated. After the addition of the above attention module, the attention-based bottleneck structure is obtained, which is used to replace the original bottleneck structure in ResNet. Finally, using features extracted from the attention-based ResNet network, a 1-D feature vector is first obtained through global average pooling, and then several layers of fully connected networks are used to obtain the result of two classifications, which are used to determine whether the input image is defective.

At present, most attention mechanisms use a fully connected structure to calculate channel weights, which increases the amount of calculation and the number of parameters of the model. In practical applications, the fabric defect detection task has specific requirements for model's efficiency and lightness. To meet the needs of actual fabric defect detection, we use a lightweight attention module [34]. In this module, the fully connected layer is replaced with the calculation of 1-D convolution, which can significantly reduce the amount of calculation and the number of parameters caused by the calculation of the attention mechanism. Its structure is shown in Fig. 4. First, the feature map is pooled, and a 1-D vector containing the global information of each channel is obtained. After that, 1-D convolution calculation and the subsequent sigmoid activation layer are used to obtain the weight corresponding to each channel. The weight is multiplied by the corresponding channel to realize the weighted calculation of information about each channel, that is, to realize filtering based on the attention of the channel domain.

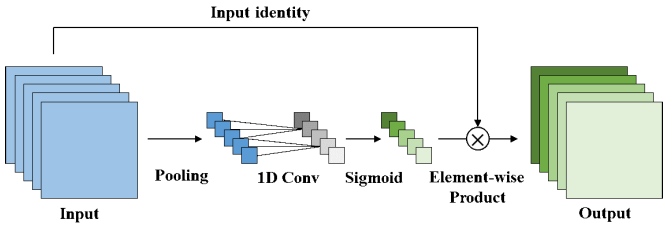


Fig. 4. Structure of the efficient channel attention (ECA)-Net residual module.

IV. Experiment and Discussion

For the defect detection model used in the experiments, the feature extraction portion of the output network was a 1-D vector with a size of 1000. The input size of the first fully connected layer was 1000, the output size was 100, the input size of the second fully connected layer was 100, and the output size was 2. The output results were used to determine whether a fabric image contained defects. We compared our tactile method with the visual method using the same ResNet model as the feature extracting layers. In the follow-up attention mechanism experiments, ResNet with an attention module based on the channel domain was used as the feature extraction model for the experiments. Transfer learning was employed in the training [35]. All parts of the CNN for extracting feature parameters were used in a pretrained ImageNet, whereas other parts of the detection model were used to initialize the parameters. Subsequently, the entire model was fine-tuned on our dataset to train the defect detection model. A separate model was used in each tactile and visual datasets for training and accurate detection of the corresponding defect to compare the overall performances of the two detection methods. For visual datasets, the introduction of RGB channel information may introduce redundant information, making it more difficult for the network to learn accurate features corresponding to defects, resulting in fluctuations in training. To avoid the influence of fluctuations in color channels of training visual images, grayscale experimental visual images were added for comparison.

In the experiment, a method termed "cross type validation" was used to ensure the credibility of the experimental results. When testing the model's detection accuracy for each fabric type, the fabric type was used as the test set, and the other eight fabric types were used as the training set. As a result, when a trained model is applied to the test set, the test set would contain new fabric types unfamiliar to the model, thus the model's robustness can be effectively verified for fabric defect detection. On this basis, the average detection accuracy

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TABLE I
DETECTION ACCURACY ON DIFFERENT FABRIC TYPES

Type	Accuracy (%)		
	Tactile	Visual	Visual (gray-level)
1	**99.69**	85.85	88.10
2	**99.84**	51.90	54.00
3	96.66	91.35	**97.93**
4	**98.96**	94.90	93.58
5	**100.00**	92.63	94.10
6	98.74	99.40	**99.73**
7	**99.85**	93.65	93.38
8	**99.93**	69.68	73.98
9	**98.21**	96.13	95.03

of all nine fabric types obtained using the cross type validation method was used as the overall detection accuracy of the model to evaluate detection performance. In addition, the difference in the detection accuracy of different types of defective fabrics was analyzed as the test set. In such experiments, the detection and recognition capabilities of various detection methods for different types of fabric defects can be observed, which can further help us analyze the detection performance.

All experiments were conducted on a workstation with an Intel Core i7-10700 CPU and a NVIDIA GeForce GTX 1080 Ti. During the training, the Adam optimizer was used and the initial learning rate was set to 0.001. The batch size was set to 8. Because transfer learning was employed, the max epoch was fixed to 25. In addition to the experimental detection efficiency, the pixel density of fabric used for detecting image pixels was 10 000 pixels/cm^2.

A. Accuracy Test Results

We performed a specific result analysis for the accuracy on each fabric type. We analyzed the differences between the tactile method and the visual method for different types of fabrics. The following results come from the model using ResNet18 without attention mechanism as the feature extracting layers.

Table I shows that regardless of training fluctuations caused by the color channel, the visual method is less accurate than the tactile method in detecting most fabric types. Among them, when fabrics 2 and 8 were used as the test set, the accuracy of the visual method was significantly lower than that of the tactile method. This is because there are irregular dyed patterns on these fabrics, and these patterns will severely affect the model's defect detection performance. Tactile information obtained by the tactile sensor was unaffected by the irregular dyed patterns, and defects in the detected image were visible. However, on some fabrics, the performance of the tactile method is not improved compared with the visual method. Type 6 is a typical fabric type for which tactile methods perform poorer than visual methods. The detection accuracy of the visual method (gray-level) is 99.73%, while the detection accuracy of the corresponding tactile method is only 98.74%. On the one hand, there are no patterns on the fabric that would interfere with visual defect detection. Since the color of the defect is white and the fabric color is purple, as shown in Fig. 2, the contrast is obvious in the gray-level visual images. The factors above make the visual method suitable for detecting the type of defect. On the other hand, the texture in the tactile images of the fabric is also very rough, which affects the detection efficiency of the tactile method. The problem can be solved with the attention mechanism.

B. Analysis of the Effect of the Attention Mechanism

From the experimental results presented in Table II, the detection accuracy of the tactile method decreases as the number of network layers increases. This is because deep networks are more susceptible to coarser texture information in tactile methods. Therefore, in this experiment, a shallow network could obtain better results. However, for practical applications, if necessary training on larger defect datasets is required, a deep network is needed. The use of the attention mechanism based on the channel domain can effectively filter fabric texture information and alleviate the impact of this problem.

The training on the tactile dataset was repeated ten times using the ResNet model with and without the attention mechanism. This is to observe the effect of the attention mechanism on the accuracy of the model training results and the stability of the training process. The training results are presented in Table II. It shows that the use of the attention mechanism can effectively improve the performance of the model, and the deeper the network model, the more significant the improvement in the accuracy of the result after using the attention mechanism. Furthermore, the standard deviation of the results of each model is reduced after the attention mechanism is introduced, which shows that the attention mechanism can make the training more stable to a certain extent.

The detection situation of each fabric type in the experiment was further analyzed before and after the introduction of the attention mechanism. Here, the results of the ResNet50 model are taken as an example. The detection results before and after the introduction of the attention mechanism are shown in Table III. After the introduction of the attention mechanism, the detection accuracy on most fabrics has improved. Besides, the attention mechanism can significantly improve the detection accuracy of specific fabrics.

In the detection results of the ResNet50 model that did not use the attention mechanism, when fabrics 6 and 9 were used as the test set for detection, the accuracy rate obtained was significantly lower than the detection accuracy rate obtained for the other seven fabric types. In the tactile images of fabrics 6 and 9, texture information is relatively coarse and irregular. The detection accuracy of the model for the two types of fabric types is lower than that of other fabric types due to the interference introduced by such texture information. After using the attention mechanism, even though the model's detection accuracy for the two fabric types was still lower than that of other fabric types, the detection accuracy was significantly improved compared to that before using the attention mechanism. Therefore, the introduction of the attention mechanism helps solve the problems caused by the tactile fabric texture information.

The influence of the attention mechanism was further investigated. Four different layers were selected from the

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TABLE II
RESULTS OF THE TEN REPETITIVE EXPERIMENTS WITH DIFFERENT DEPTH NETWORKS WITH OR WITHOUT THE ATTENTION MECHANISM

Model	Accuracy (%)		StdDev (%)	
	Attention	without Attention	Attention	without Attention
ResNet18	**99.18**	99.10	**0.127**	0.157
ResNet34	**98.49**	97.62	**1.537**	1.660
ResNet50	**98.72**	97.53	**0.990**	2.250
ResNet101	**98.83**	97.01	**0.382**	1.775
ResNet152	**97.89**	95.66	**1.065**	2.163

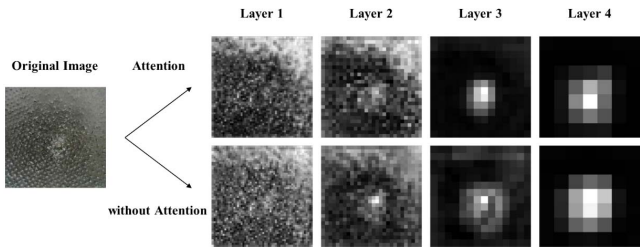


Fig. 5. Comparison of the feature map results at different layers when the model detects fabric 9 with or without the attention mechanism.

trained CNN model, and the network intermediate value was outputted during the defect detection process at these positions. After compressing the channel with an intermediate value by averaging feature map information at different layers was obtained. With the help of feature map information, we could observe the activation of different positions in the input detection image at different layers during the model detection process. Based on this information, we examined where the model pays more attention to the image during the detection process to determine the influence of the attention mechanism on the model.

Taking fabric 9 as an example, the feature maps at different layers obtained during the detection process before and after the introduction of the attention mechanism to the model are shown in Fig. 5. The whiter the color at a location, the larger the value of the feature map at that location and the stronger the activation. After introducing the attention mechanism, the activation of the fabric texture was generally lower than that before introducing the attention mechanism. This phenomenon is evident in the feature map of Layer 1 and 3, weakening the overall activation of the texture in the background. As a result, other features in the image, such as the defect features are relatively enhanced. The experiments show that the attention mechanism based on the channel domain helps the model effectively filter tactile texture information in the background, thereby improving the detection performance of the model.

Based on the above experiments, we can conclude that the introduction of the attention mechanism can effectively alleviate the influence of tactile information about the fabric texture in a tactile image on defect detection performance, and therefore significantly improve the detection accuracy of the tactile method.

C. Evaluation of the Detection Efficiency

Because the fabric defect detection method is proposed for practical applications in the fabric production process, its detection efficiency is an important factor that cannot be ignored.

TABLE III
EFFECT OF THE ATTENTION MECHANISM ON THE DETECTION RATE OF DIFFERENT FABRIC TYPES

Type	Attention	without Attention
1	**99.77**	99.31
2	99.84	**99.95**
3	**96.96**	96.51
4	**99.20**	98.93
5	**100.00**	99.84
6	**95.96**	90.40
7	99.79	**99.82**
8	**99.63**	99.50
9	**97.31**	93.50

The detection efficiency of the fabric defect detection system has two influencing factors: the data collection efficiency and the computational efficiency. In practice, the overall efficiency of the fabric defect detection system will be limited by the smaller one of the two efficiencies since a defect detection system runs the detection algorithm while collecting the data. For the efficiency of data collection, the visual method has absolute advantage over the tactile method because the tactile method requires physical contact with the fabric. For the computational efficiency, which is the main focus of this article, the efficiencies of the two methods are evaluated and compared by using the same model receiving visual and tactile images, respectively.

To measure the computational efficiency, the pixel density of a fabric detection image was first defined. Its calculation formula is as follows:

$$D = N/A \tag{1}$$

where D represents the pixel density for the fabric detection image, N represents the number of pixels used to present the fabric to be detected in the fabric detection image, and A represents the corresponding area of the fabric to be detected in reality.

When the detection algorithm and equipment are fixed, the number of image pixels that can be detected per second is fixed. Therefore, the pixel density of the fabric detection image is low, and a large fabric area can be detected simultaneously. However, the image pixel density cannot be reduced blindly because a reduction in image pixel density will affect the detection accuracy of the algorithm. Based on these assumptions, experiments were conducted to study the change in the detection accuracy under different pixel densities.

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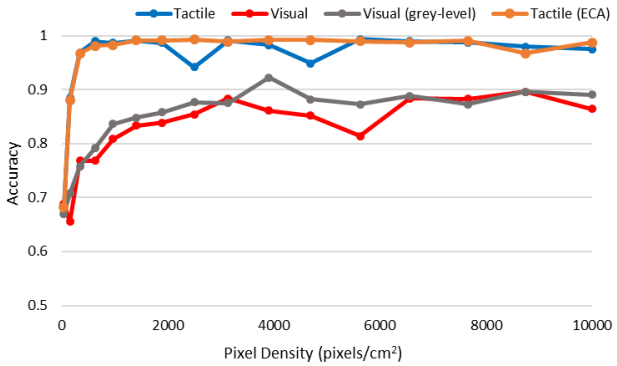


Fig. 6. Relationships between the detection accuracy and pixel density of different fabric defect detection methods.

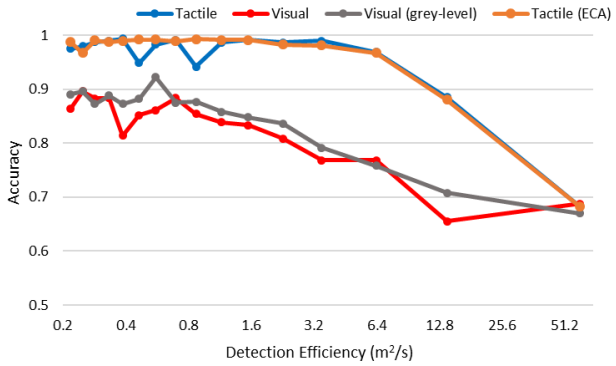


Fig. 7. Relationships between the detection accuracy and theoretical detection efficiency of different fabric defect detection methods.

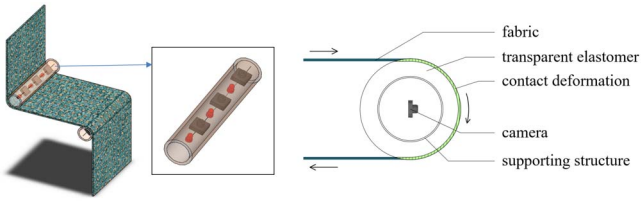


Fig. 8. Structure of a roller-type tactile detection device.

The purpose is to determine the maximum detection efficiency that each detection method can achieve while maintaining detection accuracy. In the experiment, the pixel density of the image was changed by adjusting the resolution of the fabric detection image, indirectly changing the number of pixels in the image. The theoretical detection efficiency based on the pixel density was roughly converted. It represents the maximum detection efficiency that a detection algorithm can achieve on the detection equipment under the condition that only the calculation time of the detection algorithm is considered. The theoretical detection efficiency conversion formula is as follows:

$$E = f \times n/D \tag{2}$$

where E represents the theoretical detection efficiency of the detection method, f represents the detection frame number that can be detected by the algorithm per second on the detection equipment, and n represents the number of pixels in the input frames. Thus, the detection efficiencies of the different methods can be evaluated.

Based on the ResNet50 model, fabric detection efficiency experiments were conducted. The image pixel density was changed by adjusting the image resolution. The training results of data with different pixel densities in the model are shown in Fig. 6. In addition, a rough conversion of the computational efficiency was performed according to the pixel density, and the results are shown in Fig. 7. Because only one experimental result was obtained for each image pixel density, the result will be affected by training fluctuations to a certain extent. However, in this case, using the attention method can effectively ensure the stability of training.

The results show that when the pixel density is greater than 625 pixels/cm^2, the ResNet50 model based on the attention mechanism can be used for defect detection, since the information provided by the tactile image would be sufficient to maintain the detection accuracy of the model at a high level. At this time, the computational efficiency was 3.48 m^2/s. This detection rate can meet the requirements of actual fabric production. Therefore, the use of tactile information for fabric defect detection can theoretically meet the actual demand for detection efficiency verifying the feasibility of this method.

In addition, the results show that the tactile method is more computationally efficient than the visual method because of the more obvious defect features. As we mentioned before, the overall efficiency depends on the smaller one of the data collection efficiency and the computational efficiency. As a result, as long as the data collection efficiency of the tactile method is higher than its computational efficiency, the overall detection efficiency of the tactile method can be higher than that of the visual method. The device we proposed in the next part providers a possible solution to the actual use in real-world applications.

D. Discussion

Because the size of current existing tactile sensors is typically small, the sensor needs to be constantly lifted, moved to another position and it should be pressed during the process of detecting the defects of fabrics. This action is frequent and time-consuming and becomes a time bottleneck, significantly lowering the actual detection efficiency of the tactile detection method compared to the theoretical detection efficiency. Therefore, a more efficient tactile detection device is required to adapt to the actual fabric production process. For this reason, a roller-type tactile detection device is proposed for fabric defect detection, and its structure is shown in Fig. 8. The composition of the roller-type tactile sensor is similar to that of the aforementioned tactile sensor, but the transparent elastic body and its supporting structure are designed in the shape of a roller according to the production characteristics of the fabric to improve detection efficiency. The camera is fixed on a shaft located in the center of the drum, and it always records deformation data of the elastic body toward the side at which the fabric is in contact with the drum. With this device, the tactile sensor can continuously obtain surface tactile information about the fabric produced during the production process. Because there is no need to adjust sensor positions, the device can give full play to the efficiency

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of the detection algorithm. When using this device, the actual efficiency of fabric defect detection can be comparable to the theoretical efficiency calculated in the previous experiment, which provides a possible solution to the practical application of tactile methods in fabric defect detection.

V. Conclusion

A tactile-based detection method is proposed for fabric defects. This method can avoid the influence of color information, such as the texture fabric itself and dyeing patterns on the versatility of the detection algorithm, thereby simplifying the detection algorithm. In addition, the impact of ambient light conditions on the detection process can be reduced, which will improve the robustness of the detection. Subsequently, experiments on the tactile detection method were conducted, and the results of the visual methods were employed for comparative analysis. The results verified that tactile information is unaffected by fabric surface patterns compared with visual information in the detection of structural defects. Information has obvious advantages. Aiming at the influence of rougher texture information in tactile images on the deep network model, an attention mechanism was introduced to the model. The results verify that the attention mechanism can effectively filter rough texture information and alleviate its impact on the detection model. Furthermore, it can not only effectively improve the detection accuracy of the tactile detection method but also make the training of the model more stable. In addition, we conducted experiments on the detection efficiency of each method and verified that the use of tactile information for fabric defect detection can theoretically meet requirements in term of detection efficiency in the fabric production process, thereby verifying the feasibility of the proposed method.

Although the proposed tactile sensor is more sensitive and can extract more robust feature in the detection of structural defects than traditional visual sensors, there are also some deficiencies that restrict the practical use of the tactile sensor. For example, the tactile sensor cannot collect the visual information of fabrics, thus unable to detect dyeing defects on fabrics. In the future, we will focus on the combination of visual and tactile information. This method can compensate for the inability to detect dyeing defects using only tactile information, so that a more general and robust industrial fabric inspection system for practical use can be obtained.

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